

Sum of Divisors

Time limit: 1.00 s

Memory limit: 512 MB

Let $\sigma(n)$ denote the sum of divisors of an integer n . For example, $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$.

Your task is to calculate the sum $\sum_{i=1}^n \sigma(i)$ modulo $10^9 + 7$.

Input

The only input line has an integer n .

Output

Print $\sum_{i=1}^n \sigma(i)$ modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^{12}$

Example

Input	Output
5	21